

A Leakage-Free Machine Learning Classifier for Colon Cancer Proliferation Using Cross-Platform Transcriptomic Validation

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Background

Proliferation rate is a key prognostic marker in colorectal cancer, typically measured via Ki-67 immunohistochemistry which suffers from inter-observer variability. A transcriptomic classifier trained on gene expression data could provide an objective, reproducible alternative. However, most cancer ML studies overfit to a single platform and fail to generalize across microarray and RNA-seq cohorts.

Methods

Data. Training on GEO microarray (GSE39582, n=585). Independent external validation on TCGA-COAD RNA-seq (n=322) and CPTAC-COAD RNA-seq (n=105).

Labeling. Proliferation state defined by median split of the 10-gene proliferation signature (CIN70 subset). These 10 genes were removed from all feature matrices before training to prevent target leakage.

Pipeline. StandardScaler, VarianceThreshold, and SelectKBest encapsulated in an sklearn Pipeline with parameters learned on training folds only. Models: Logistic Regression (L2), Random Forest, XGBoost, MLP. Platt scaling applied for cross-platform calibration.

Results

Table 1. Best-performing model (Random Forest) across training and validation cohorts.

Cohort	N	Platform	ROC-AUC	Accuracy	Brier
GEO (training)	585	Microarray	0.978	0.912	0.077
TCGA-COAD	322	RNA-seq	0.973	0.921	0.065
CPTAC-COAD	105	RNA-seq	0.949	0.868	0.096

Survival analysis. Kaplan-Meier log-rank test significant in both GEO (p = 0.037) and TCGA (p = 0.034). Multivariate Cox PH: proliferation status showed a trend toward association (HR = 0.78, p = 0.092) but did not reach statistical significance. CPTAC-COAD log-rank test was underpowered (p = 0.356, only 7 survival events).

Conclusions

The pipeline achieves cross-platform ROC-AUC exceeding 0.95 on two independent RNA-seq validation cohorts despite training exclusively on microarray data. This level of generalizability, enabled by rigorous leakage prevention and Platt calibration, indicates that the classifier captures genuine biological signal rather than platform-specific artifacts. The proliferation classifier offers a reproducible, objective alternative to Ki-67 immunohistochemistry and warrants further validation in prospective cohorts.